

Customer Segmentation & Business Intelligence Analysis

Report Overview

Report title	Customer Segmentation & Business Intelligence Analysis
Programme	ApexPlanet Data Analytics Internship — Task 3 (Deep-Dive Analysis)
Dataset name	ApexPlanet_Cleaned_Dataset.xlsx
Dataset size	1,000 rows × 12 columns
Unique customers	947 (unique Customer_ID values)
Unique Order IDs	992
Analysis period	Jan 2025 to Jan 2026 (monthly sales output)
Total sales analysed	139,399,439.65

Task 3 uses the same cleaned dataset as Task 1 and Task 2, because no separate dataset was provided. All figures in this report are taken directly from the uploaded analysis output files; no values have been estimated, modified or replaced. Currency values are shown as plain numbers with comma separators and two decimal places.

2. Executive Summary

The dataset records total sales of **139,399,439.65** across **992** unique Order IDs, **947** unique customers and **5,435** units sold. Average Order Value is 140,523.63 and Sales per Customer is 147,201.10.

- **Customer segmentation.** Using the 25th and 75th percentiles of customer-level Total Sales, the 947 customers split into 237 High Value, 473 Medium Value and 237 Low Value customers.
- **Revenue concentration.** High Value customers are 25.03% of customers but generate 55.27% of sales (77,048,925.29). Low Value customers are 25.03% of customers and contribute 4.04%.
- **Category performance.** Electronics is the leading category with 50,778,581.70 (36.43% of sales), roughly twice the next category, Education (17.96%).
- **Geographic performance.** Patna records the highest total sales (19,285,966.89), while Bengaluru records the highest sales per customer (156,446.45). These are different metrics and lead to different rankings.
- **Monthly trend.** Among full months, Mar 2025 is the highest (13,059,899.94) and Sep 2025 the lowest (9,179,896.29). Jan 2026 contains only 809,389.99 and should not be compared with complete months.
- **Data quality.** Order ID ORD100050 appears on 9 records belonging to 9 different customers, dates and products. These are treated as separate transactions, and the repeated Order ID is flagged as a data-quality issue. Thirteen records carry the city value “Unknown”.
- **Opportunities.** The main opportunities are retaining the High Value segment, developing the large Medium Value base and the low-spending Low Value segment, growing categories beyond Electronics, and building an interactive Power BI dashboard for ongoing monitoring.

3. Dataset Validation

The table below summarises the validation figures available in the uploaded analysis outputs.

Validation item	Result
Number of rows	1,000
Number of columns	12 (Order_ID, Order_Date, Customer_ID, Customer_Name, Age, Gender, City, Product, Category, Quantity, Unit_Price, Total_Sales)
Unique customers (Customer_ID)	947
Unique Order IDs	992
Repeated customer rows	53
Repeated Order ID rows	8
Missing values	Not reported in the uploaded analysis outputs. The customer-level table (947 customers) contains no blank cells.
Duplicate rows (fully identical)	Not reported in the uploaded analysis outputs.
Order date range	Monthly output spans Jan 2025 to Jan 2026. Exact first and last order dates are not included in the uploaded files.

Data-quality finding: repeated Order ID

The Order ID **ORD100050** appears on nine records. As shown below, each record has a different customer, order date, product and unit price, so they are distinct transactions that share an identifier rather than duplicates of one transaction. They have therefore not been merged or removed. Order ID should not be used as a unique key in this dataset; Customer_ID is the reliable customer key.

Order ID	Order Date	Customer ID	City	Product	Qty	Total Sales
ORD100050	2025-10-03	CUST5523	Pune	Chair	9	277,513.83
ORD100050	2025-01-22	CUST3194	Kolkata	Book	2	51,591.10
ORD100050	2025-03-01	CUST8467	Bengaluru	Book	3	8,582.49
ORD100050	2025-02-11	CUST6467	Pune	Rice	7	318,362.73
ORD100050	2025-11-22	CUST8225	Bengaluru	Rice	8	311,660.72
ORD100050	2025-02-21	CUST8859	Unknown	Laptop	7	141,013.88
ORD100050	2025-11-06	CUST7824	Mumbai	Rice	4	170,570.36
ORD100050	2025-08-03	CUST5585	Kolkata	Chair	9	258,727.23
ORD100050	2025-02-09	CUST7188	Bengaluru	Mobile	1	34,619.77

Table 3.1 — Records sharing Order ID ORD100050 (source: 00_Duplicate_Order_IDs.csv).

Reconciliation note. Order counts in the segment, city, category and product tables are not designed to add up to 992 Unique Order IDs, because ORD100050 is counted in each group where it appears (for example, in 5 cities and 5 products). Customer-level Total_Orders add up to 1,000 records. Average Order Value in the Core KPIs is calculated on the 992 Unique Order IDs, as supplied.

Another observation: 13 orders (1,540,617.54 in sales) are recorded under the city value “Unknown”, which limits geographic analysis for these records.

4. Core Business KPIs

KPI	Value
Total Sales	139,399,439.65
Unique Customers	947
Unique Order IDs	992
Total Quantity	5,435
Average Order Value	140,523.63
Sales per Customer	147,201.10

Table 4.1 — Core business KPIs (source: 01_Core_KPIs.csv).

Average Order Value is Total Sales divided by Unique Order IDs. Sales per Customer is Total Sales divided by Unique Customers.

5. Customer-Level Analysis

Customer_ID is the unique customer key. Transactions were aggregated to one row per Customer_ID, giving **947 unique customers**. For each customer the analysis records Total Sales, Total Orders, Total Quantity, Average Order Value and demographic information (Customer Name, Age, Gender, City).

Customer-level Total Sales	Value
Customers (unique Customer_ID)	947
Total Sales per customer — minimum	437.34
25th percentile	47,859.02
Median	112,029.57
Mean (equals Sales per Customer)	147,201.10
75th percentile	214,435.46
Total Sales per customer — maximum	867,333.24

Table 5.1 — Distribution of customer-level Total Sales (calculated from 02_Customer_Analysis.csv).

Order frequency and demographics

- Orders per customer: 895 customers have 1 order, 51 have 2 orders and 1 has 3 orders; most customers therefore appear once in the data.
- Age ranges from 18 to 65 years (mean 41.3).
- Gender: 463 female and 484 male customers.

6. Customer Segmentation

Methodology

Customers were ranked by customer-level Total Sales and classified using the 25th and 75th percentiles of that distribution:

- **Low Value:** Total Sales at or below the 25th percentile (47,859.02).
- **Medium Value:** Total Sales above the 25th percentile and up to the 75th percentile (214,435.46).
- **High Value:** Total Sales above the 75th percentile.

Customer Segment	Customers	Customer %	Total Sales	Sales Contribution %
High Value	237	25.03%	77,048,925.29	55.27%
Medium Value	473	49.95%	56,722,617.47	40.69%
Low Value	237	25.03%	5,627,896.89	4.04%
Total	947	100.00%	139,399,439.65	100.00%

Table 6.1 — Customer segment summary (source: 03_Segment_Summary.csv). Segment customer counts reconcile to 947.

Customer Segment	Total Orders	Total Quantity	Average Order Value
High Value	270	2,082	296,265.28
Medium Value	491	2,590	117,155.35
Low Value	239	763	23,676.47

Table 6.2 — Orders, quantity and Average Order Value by segment (source: 03_Segment_Summary.csv). Segment Average Order Value is the average of customer-level values as supplied.

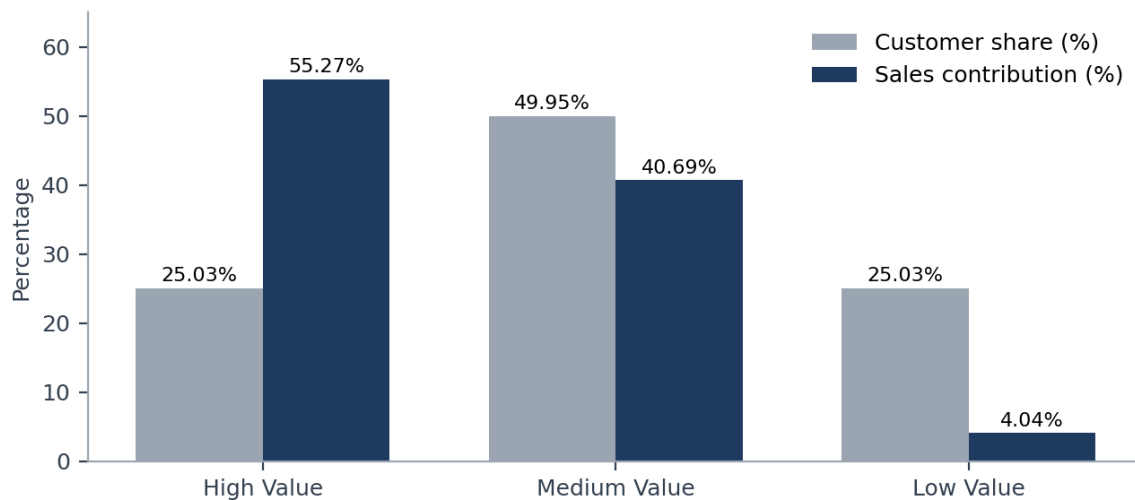


Figure 6.1 — Customer share versus sales contribution by segment.

7. Customer Segment Insights

- **Revenue concentration.** The 237 High Value customers account for 55.27% of total sales from 25.03% of customers. Over half of all revenue depends on one quarter of the customer base.
- **High Value importance.** High Value customers have an Average Order Value of 296,265.28, compared with 117,155.35 for Medium Value and 23,676.47 for Low Value customers.
- **Low Value opportunity.** Low Value customers equal the High Value segment in size (237 customers) but contribute only 5,627,896.89 (4.04%). Any upward movement in this group would come from a very low base.
- **Segment distribution.** Medium Value is the largest segment with 473 customers (49.95%) and contributes 40.69% of sales, so it represents the largest pool of customers who could move between segments.
- **Spend ranges.** Customer-level Total Sales ranges from 437.34 to 47,848.00 for Low Value, 47,870.04 to 213,995.10 for Medium Value, and 214,875.81 to 867,333.24 for High Value.

8. Top 10 Customers

Customer ID	Customer Name	Age	Gender	City	Total Sales	Order s	Qty	Segment
CUST9510	Customer_345	21	Female	Kolkata	867,333.24	2	19	High Value
CUST6845	Customer_337	34	Male	Hyderabad	769,479.75	2	18	High Value
CUST6532	Customer_348	30	Male	Pune	610,981.74	2	15	High Value
CUST6082	Customer_301	62	Male	Mumbai	548,416.39	2	16	High Value
CUST3689	Customer_179	44	Female	Delhi	542,484.08	2	16	High Value
CUST3730	Customer_273	41	Female	Unknown	538,633.97	2	13	High Value
CUST4706	Customer_34	62	Female	Delhi	528,036.44	2	11	High Value
CUST9693	Customer_343	48	Male	Patna	511,421.86	2	13	High Value
CUST7374	Customer_375	62	Female	Patna	511,309.40	2	15	High Value
CUST2062	Customer_254	58	Male	Patna	493,677.50	1	10	High Value

Table 8.1 — Top 10 customers by Total Sales (source: 04_Top_10_Customers.csv).

- All ten customers belong to the High Value segment. Together they account for 5,921,774.37 in sales, which is 4.25% of total sales.
- Nine of the ten customers have two orders; the tenth has one. The highest customer has a Total Sales of 867,333.24.
- Because the top ten represent a small share of total sales, High Value concentration is a segment-wide pattern, rather than dependence on a handful of individual customers.

9. Segment × Gender Analysis

Customer Segment	Female	Male	Total	Female %	Male %
High Value	106	131	237	44.73%	55.27%
Medium Value	240	233	473	50.74%	49.26%
Low Value	117	120	237	49.37%	50.63%
Total	463	484	947	48.89%	51.11%

Table 9.1 — Customer count and percentage by segment and gender (sources: 05_Segment_Gender.csv, 06_Segment_Gender_Percentage.csv). Percentages are within each segment.

- Overall the customer base is close to balanced: 463 female and 484 male customers.
- High Value has a higher male share (55.27%) than female share (44.73%); Medium Value is slightly female-leaning (50.74%); Low Value is almost evenly split.
- The uploaded outputs provide customer counts by gender only. They do not include sales by gender, so no conclusion about spending behaviour by gender is drawn here.

10. Segment × City Analysis

City	High Value	Medium Value	Low Value	Total Customers
Bengaluru	34	55	27	116
Delhi	31	54	32	117
Gaya	20	61	31	112
Hyderabad	27	58	34	119
Kolkata	33	67	29	129
Mumbai	31	65	27	123
Patna	32	66	29	127
Pune	27	42	23	92
Unknown	2	5	5	12
Total	237	473	237	947

Table 10.1 — Customers by city and segment (source: 07_Segment_City.csv). Each customer is assigned to one city; the total reconciles to 947.

- Kolkata has the most customers (129) and the most Medium Value customers (67).
- Bengaluru has the most High Value customers (34), followed by Kolkata (33) and Patna (32).
- Gaya has the fewest High Value customers among named cities (20) and a comparatively large Low Value group (31).
- Pune has the fewest customers among named cities (92). The “Unknown” city holds 12 customers.

11. Segment × Category Analysis

Category	High Value	Medium Value	Low Value	Total
Electronics	28,681,180.05	20,088,556.06	2,008,845.59	50,778,581.70
Education	13,490,462.28	10,620,090.60	921,136.52	25,031,689.40
Grocery	13,654,685.92	7,715,683.58	861,341.78	22,231,711.28
Furniture	12,011,083.02	8,667,797.67	842,680.79	21,521,561.48
Fashion	9,211,514.02	9,630,489.56	993,892.21	19,835,895.79
Total	77,048,925.29	56,722,617.47	5,627,896.89	139,399,439.65

Table 11.1 — Sales by category and customer segment (source: 08_Segment_Category.csv).

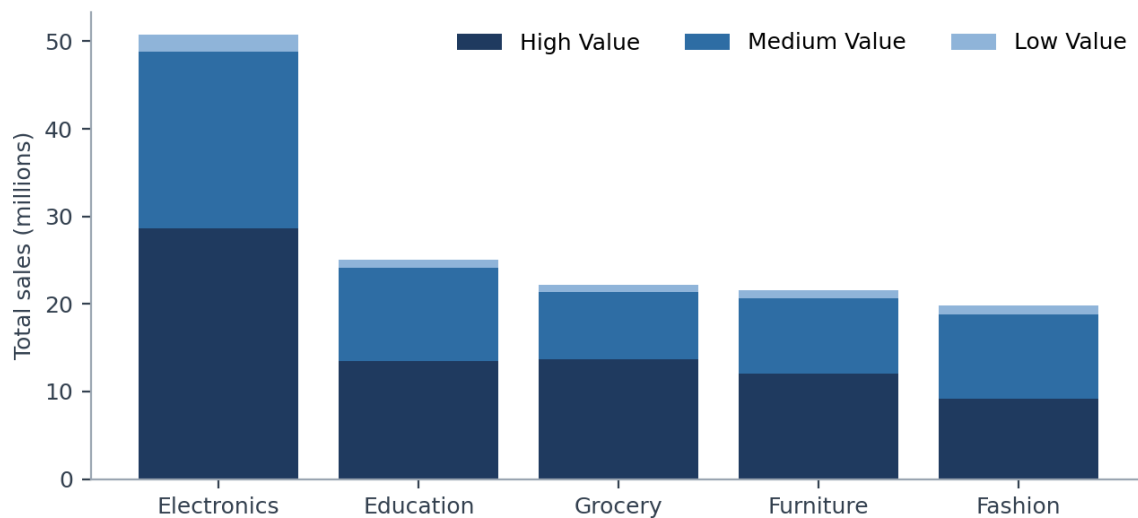


Figure 11.1 — Sales by category, split by customer segment.

- **High Value:** Electronics is the largest category (28,681,180.05), followed by Grocery (13,654,685.92) and Education (13,490,462.28).
- **Medium Value:** Electronics is the largest (20,088,556.06), followed by Education (10,620,090.60) and Fashion (9,630,489.56).
- **Low Value:** Electronics is the largest (2,008,845.59), followed by Fashion (993,892.21) and Education (921,136.52).
- Fashion is the only category where Medium Value sales (9,630,489.56) exceed High Value sales (9,211,514.02). High Value share is highest in Grocery (61.42%) and lowest in Fashion (46.44%) (calculated from Table 11.1).

12. Segment × Product Analysis

Product	High Value	Medium Value	Low Value	Total
Laptop	15,828,557.45	8,584,864.35	1,029,586.71	25,443,008.51
Mobile	12,852,622.60	11,503,691.71	979,258.88	25,335,573.19
Book	13,490,462.28	10,620,090.60	921,136.52	25,031,689.40
Rice	13,654,685.92	7,715,683.58	861,341.78	22,231,711.28
Chair	12,011,083.02	8,667,797.67	842,680.79	21,521,561.48
Shoes	9,211,514.02	9,630,489.56	993,892.21	19,835,895.79
Total	77,048,925.29	56,722,617.47	5,627,896.89	139,399,439.65

Table 12.1 — Sales by product and customer segment (source: 09_Segment_Product.csv).

- Laptop is the largest product for High Value customers (15,828,557.45), with High Value customers providing 62.21% of Laptop sales.
- Mobile is the largest product for Medium Value customers (11,503,691.71), and High Value customers provide 50.73% of Mobile sales, the most balanced mix among the Electronics products.
- Shoes is the only product where Medium Value sales (9,630,489.56) exceed High Value sales (9,211,514.02). Shoes is also the only product in the Fashion category, so the Fashion pattern is the same as the Shoes pattern.
- Each category other than Electronics contains a single product, so the category and product tables show identical values for Book, Chair, Rice and Shoes.
- Percentage shares in this section are calculated from Table 12.1.

13. Category Performance

Category	Total Sales	Total Orders	Total Quantity	Sales Contribution %
Electronics	50,778,581.70	353	1,978	36.43%
Education	25,031,689.40	177	977	17.96%
Grocery	22,231,711.28	151	826	15.95%
Furniture	21,521,561.48	158	855	15.44%
Fashion	19,835,895.79	156	799	14.23%

Table 13.1 — Category performance (source: 13_Category_Summary.csv). Orders are counted per category using distinct Order IDs.

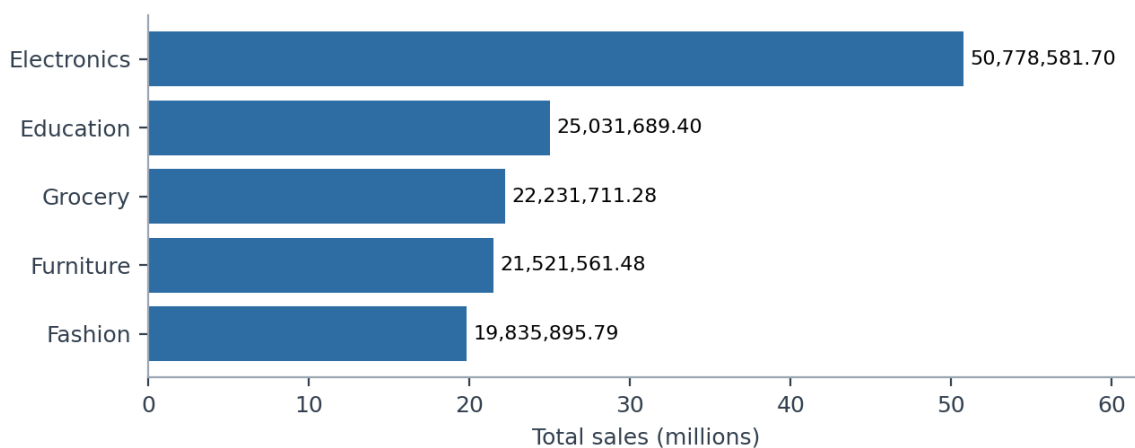


Figure 13.1 — Total sales by category.

Electronics is the highest-performing category, with 50,778,581.70 in sales, 353 orders and 1,978 units (36.43% of sales). The other four categories are closer together, from 17.96% (Education) to 14.23% (Fashion). Electronics includes two products (Laptop and Mobile), whereas each other category has one product.

14. Product Performance

Product	Total Sales	Total Orders	Total Quantity	Sales Contribution %
Laptop	25,443,008.51	170	970	18.25%
Mobile	25,335,573.19	184	1,008	18.17%
Book	25,031,689.40	177	977	17.96%
Rice	22,231,711.28	151	826	15.95%
Chair	21,521,561.48	158	855	15.44%
Shoes	19,835,895.79	156	799	14.23%

Table 14.1 — Product performance (source: 14_Product_Summary.csv).

- The top three products are Laptop (25,443,008.51; 18.25%), Mobile (25,335,573.19; 18.17%) and Book (25,031,689.40; 17.96%). Their sales are within about 0.30 percentage points of each other.
- Laptop leads Mobile by 107,435.32 in sales, but Mobile has more orders (184 vs 170) and more units (1,008 vs 970).
- Shoes has the lowest sales (19,835,895.79; 14.23%).

15. City Performance

City	Total Sales	Orders	Quantity	Customers	Sales per Customer
Patna	19,285,966.89	135	753	134	143,925.13
Kolkata	18,884,349.57	132	748	132	143,063.25
Bengaluru	18,773,574.32	120	670	120	156,446.45
Mumbai	18,757,050.17	131	707	130	144,285.00
Hyderabad	17,166,766.87	125	669	125	137,334.13
Delhi	16,097,079.00	125	662	124	129,815.15
Pune	14,513,175.90	98	552	99	146,597.74
Gaya	14,380,859.39	117	609	117	122,913.33
Unknown	1,540,617.54	13	65	13	118,509.04

Table 15.1 — City performance (source: 12_City_Summary.csv). Customers are distinct Customer_IDs with transactions in that city.

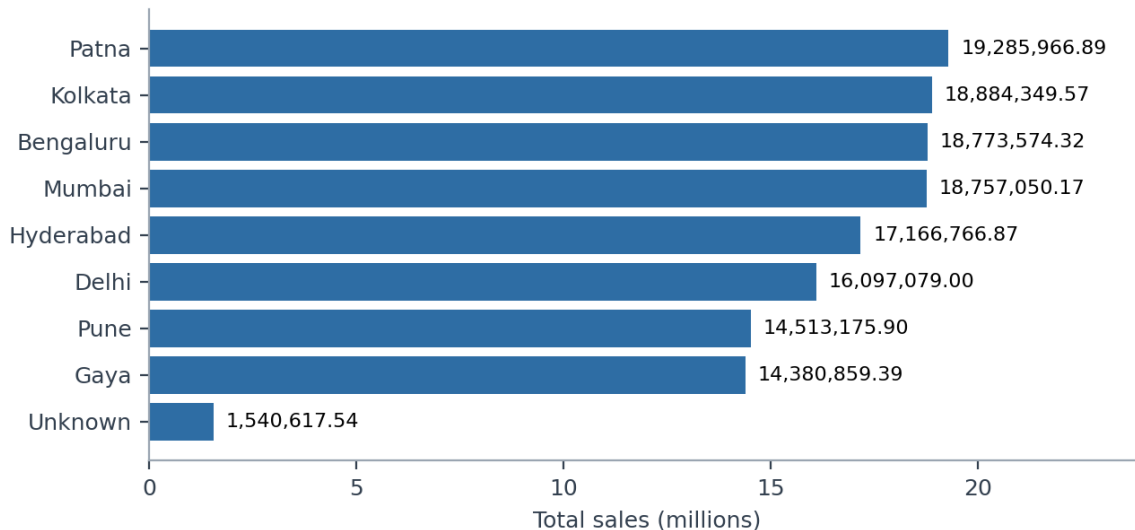


Figure 15.1 — Total sales by city.

Two different metrics

- **Highest total sales:** Patna with 19,285,966.89, followed by Kolkata (18,884,349.57) and Bengaluru (18,773,574.32).
- **Highest sales per customer:** Bengaluru with 156,446.45, ahead of Pune (146,597.74) and Mumbai (144,285.00).
- Patna leads on total sales but its sales per customer (143,925.13) is below Bengaluru, Pune and Mumbai; the two metrics should not be treated as the same.
- Gaya has the lowest sales per customer among named cities (122,913.33). The “Unknown” group has 118,509.04 per customer on only 13 customers.

Note on customer counts. The Customers column above adds up to 994, which is higher than the 947 unique customers. This is because a Customer_ID with transactions in more than one city is counted in each of those cities here, whereas Table 10.1 assigns each customer to a single city.

16. Monthly Sales Trend

Month	Total Sales	Change vs prior month	High Value	Medium Value	Low Value
Jan 2025	10,096,190.73	—	5,309,683.78	4,322,932.68	463,574.27
Feb 2025	11,511,181.46	+14.02%	5,326,018.91	5,497,525.11	687,637.44
Mar 2025	13,059,899.94	+13.45%	7,065,641.55	5,516,185.04	478,073.35
Apr 2025	12,222,700.17	-6.41%	6,636,151.35	5,365,226.98	221,321.84
May 2025	10,984,689.25	-10.13%	6,726,954.60	3,683,938.65	573,796.00
Jun 2025	12,912,332.64	+17.55%	7,911,378.60	4,574,474.88	426,479.16
Jul 2025	11,746,226.80	-9.03%	6,318,511.65	4,957,018.87	470,696.28
Aug 2025	9,448,471.22	-19.56%	4,382,827.50	4,603,387.76	462,255.96
Sep 2025	9,179,896.29	-2.84%	4,096,886.12	4,459,700.04	623,310.13
Oct 2025	12,500,936.50	+36.18%	8,144,736.85	4,031,811.58	324,388.07
Nov 2025	12,627,620.22	+1.01%	8,311,632.06	4,014,400.42	301,587.74

Month	Total Sales	Change vs prior month	High Value	Medium Value	Low Value
Dec 2025	12,299,904.44	-2.60%	6,435,481.45	5,269,646.34	594,776.65
Jan 2026	809,389.99	n/a (partial)	383,020.87	426,369.12	0.00
Total	139,399,439.65		77,048,925.29	56,722,617.47	5,627,896.89

Table 16.1 — Monthly sales and segment split (sources: 11_Monthly_Sales.csv, 10_Monthly_Segment_Sales.csv).
Month-on-month change is calculated from Total Sales.

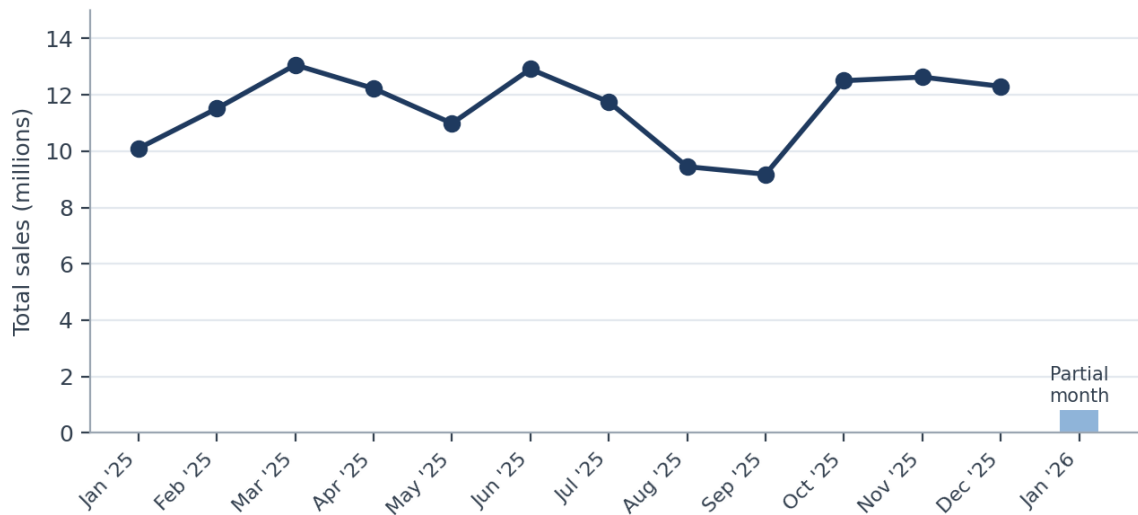


Figure 16.1 — Monthly sales trend. The final month is partial and shown separately.

Partial final month. Jan 2026 records only 809,389.99, far below every other month (all above 9,000,000), which indicates an incomplete month at the end of the data range. It should not be compared directly with complete months, and it is excluded from the highest and lowest month comparison. The exact last order date is not included in the uploaded files.

- **Highest full month:** Mar 2025 (13,059,899.94). **Lowest full month:** Sep 2025 (9,179,896.29).
- **Major peaks:** Mar 2025 (13,059,899.94), Jun 2025 (12,912,332.64), Nov 2025 (12,627,620.22) and Oct 2025 (12,500,936.50).
- **Major declines:** Aug 2025 fell 19.56% from July, and May 2025 fell 10.13% from April. Sales then rose 36.18% in Oct 2025.
- In August and September 2025, High Value sales were 4,382,827.50 and 4,096,886.12, the lowest High Value totals among full months, which coincides with the overall dip. In October and November, High Value sales reached 8,144,736.85 and 8,311,632.06, the highest among full months.

17. Key Business Insights

Insight 1: Revenue is concentrated in the High Value segment

Finding: 237 customers (25.03%) generate 55.27% of sales (77,048,925.29).

Business meaning: A relatively small group of customers drives most revenue.

Why it matters: Losing even a part of this group would have a large effect on total sales, so retaining it is a priority.

Insight 2: Low Value customers contribute very little

Finding: 237 customers (25.03%) contribute 4.04% of sales, with an Average Order Value of 23,676.47.

Business meaning: This segment is as large as the High Value segment but spends at a much lower level.

Why it matters: It marks where growth in order value could be explored, although it starts from a low base.

Insight 3: Electronics dominates category sales

Finding: Electronics generates 50,778,581.70 (36.43% of sales), about twice the next category, Education.

Business meaning: Revenue is weighted towards one category, although it combines two products (Laptop and Mobile).

Why it matters: Performance of Electronics has a stronger effect on total sales than any other category.

Insight 4: Top products are closely matched

Finding: Laptop (18.25%), Mobile (18.17%) and Book (17.96%) are within a narrow range; Mobile has more orders and units than Laptop.

Business meaning: Product leadership depends on the measure used: Laptop on sales value, Mobile on order and unit volume.

Why it matters: Product decisions should consider both value and volume.

Insight 5: Total sales and sales per customer rank cities differently

Finding: Patna has the highest total sales (19,285,966.89); Bengaluru has the highest sales per customer (156,446.45).

Business meaning: Patna is a high-volume city, while Bengaluru has higher value per customer.

Why it matters: Geographic strategy may differ for scale versus customer value.

Insight 6: Sales are uneven across months

Finding: Full-month sales range from 9,179,896.29 (Sep 2025) to 13,059,899.94 (Mar 2025), with a dip in August and September 2025 and a recovery in October.

Business meaning: Monthly sales are not stable, and High Value sales move with the overall pattern.

Why it matters: Regular monthly monitoring helps identify slow periods early.

Insight 7: Fashion has a different customer mix

Finding: Fashion (Shoes) is the only category where Medium Value sales (9,630,489.56) exceed High Value sales (9,211,514.02).

Business meaning: Fashion relies more on mid-spending customers than other categories do.

Why it matters: Category plans may need segment-specific approaches.

Insight 8: Data-quality issues affect some measures

Finding: ORD100050 is shared by nine different transactions, and 13 records have the city “Unknown”.

Business meaning: Order IDs cannot be used as unique transaction keys, and some geographic analysis is incomplete.

Why it matters: Improving identifiers and city capture would improve the reliability of future order and city analysis.

18. Business Recommendations

The following recommendations are based on the findings above. They are suggestions for evaluation, and no outcome is guaranteed.

High Value customer retention	Consider a dedicated retention approach for the 237 High Value customers, such as account reviews or tailored offers, and track whether repeat purchasing changes. Monitor the segment's monthly sales, particularly in periods like August and September 2025 when High Value sales were lower.
Low Value customer development	Test low-cost actions, such as bundles or product recommendations, aimed at raising order value among Low Value customers, and measure the results by segment movement over time.
Category growth	Review options to grow categories beyond Electronics (Education, Grocery, Furniture, Fashion), and examine why Fashion relies more on Medium Value customers.
Product strategy	Compare Laptop and Mobile on both value and volume. Review Shoes, the lowest-selling product, to understand whether pricing, range or promotion changes should be tested.
Geographic strategy	Treat total sales and sales per customer as separate goals: consider volume-building ideas for cities with lower totals such as Gaya and Pune, and customer-value ideas where sales per customer is lower, such as Gaya and Delhi.
Monthly performance monitoring	Track monthly sales and the segment split on a regular dashboard, compare only complete months, and investigate the August-September 2025 dip with additional data.
Data quality improvement	Enforce unique Order IDs, record city information at the point of sale so “Unknown” values are reduced, and add checks for missing values and duplicates to the data-cleaning process.

19. Power BI Dashboard Requirements

The findings in this report can be converted into an interactive Power BI dashboard. The tables below describe a recommended layout, with the uploaded analysis output that could feed each element.

Dashboard element	Content	Source output
KPI Cards	Total Sales, Unique Customers, Unique Order IDs, Total Quantity, Average Order Value, Sales per Customer	01_Core_KPIs.csv
Customer Segment Distribution	Customers and Customer % by segment (donut or bar)	03_Segment_Summary.csv
Sales Contribution by Segment	Total Sales and Sales Contribution % by segment	03_Segment_Summary.csv
Monthly Sales Trend	Line chart of monthly sales with segment split; mark the partial final month	11_Monthly_Sales.csv, 10_Monthly_Segment_Sales.csv
Sales by Category	Bar chart with segment breakdown	13_Category_Summary.csv, 08_Segment_Category.csv
Sales by City	Bar chart or map, with sales per customer as a second measure	12_City_Summary.csv
Top 10 Customers	Ranked table with segment	04_Top_10_Customers.csv
Segment × Product Matrix	Matrix with products as rows and segments as columns	09_Segment_Product.csv

Table 19.1 — Recommended dashboard elements.

Slicer	Purpose
Date	Filter by month or date range; the partial final month should be clearly labelled
City	Filter all visuals by city, including “Unknown”
Gender	Filter by Female or Male
Category	Filter by product category
Product	Filter by individual product
Customer Segment	Filter by High, Medium or Low Value

Table 19.2 — Recommended slicers.

Modelling notes

- Use Customer_ID as the customer key; do not use Order_ID as a unique key because ORD100050 is repeated.
- Base the segment slicer on the segment labels in the customer-level table so segment counts stay consistent with 947 customers.
- Label order counts carefully, since distinct Order IDs within a category, product or city will not add up to 992.
- A transaction-level table connected to the customer-level table would allow all slicers to filter every visual.

20. Conclusion

The analysis of 947 unique customers and total sales of 139,399,439.65 shows that revenue is concentrated among a minority of customers: High Value customers (25.03% of customers) contribute 55.27% of sales, while Low Value customers contribute 4.04%. Segment counts and sales reconcile to the customer and sales totals in the Core KPIs.

Electronics is the leading category (36.43% of sales), with Laptop and Mobile the two largest products. Geographically, Patna has the highest total sales and Bengaluru the highest sales per customer. Monthly sales vary between 9,179,896.29 and 13,059,899.94 across full months, with Jan 2026 being a partial month that cannot be compared with complete months.

Data-quality observations include the repeated Order ID ORD100050 across nine different transactions and 13 records with an “Unknown” city. These do not change the main findings but should be addressed to improve future analysis.

An interactive Power BI dashboard would let decision-makers explore these results by segment, category, product, city, gender and date, and would support ongoing monitoring of customer value and sales performance.